Your VLA is Secretly kNN: A Training-Free Probe for Task Difficulty, Backbones and Action Heads

Anonymous Author(s)
Affiliation
Address
email

Abstract: Vision-language-action (VLA) models now exceed 95% success on dense simulation benchmarks such as LIBERO, but it is unclear what such numbers measure. We use a training-free k-nearest-neighbour lookup over the demonstrations, not as a controller but as a probe. On LIBERO it solves 91.7% of the official evaluation without training; nine public VLAs fail on the same tasks, and a lookup predicts their action chunks as well as it predicts a held-out human demonstration. Under the perturbations of LIBERO-plus the lookup degrades least of all the policies we test, while VLAs lose up to 64.8 points. Keying the same lookup on a VLA's own vision-language model separates the backbone from the action head, before and after fine-tuning: fine-tuning mostly makes the backbone a more robust key, it can also break it, and some action heads are worse than a lookup on their own backbone. A neighbour metric computed offline tracks backbone quality without rollouts. The probe is a LeRobot policy plugin, so any benchmark LeRobot runs can run the test.

Keywords: Vision-language-action models, evaluation, nearest neighbours

1 Introduction

Vision-language-action models are fine-tuned on a few dozen demonstrations per task and evaluated in simulation, where the strongest now succeed in 97% of LIBERO episodes [1]. Such a number says little by itself: the test scenes are close to the demonstrations, so a policy can succeed by recalling them. Nor does it say which part of a VLA works, the pretrained vision-language backbone or the action head trained on top of it.

We propose a simple instrument for both questions: a k-nearest-neighbour (kNN) lookup over the demonstration frames, which executes the actions that followed the frames closest to the current observation. It is *not* a proposal for robot control: a lookup cannot extrapolate beyond its memory. That is what makes it useful as a probe. It answers three questions without training anything:

1. **How hard is the task?** What a lookup over generic image features already solves is retrieval, not generalisation (§4).
2. **How good is the backbone, and what did fine-tuning do to it?** The same lookup keyed on a VLA's own vision-language model (VLM), before and after fine-tuning, or offline, without rollouts (§5).
3. **How good is the action head?** Its margin over a lookup on its own backbone (§5).

We implement the probe as a policy plugin for LeRobot [2]: it is fitted from any LeRobot dataset with one command and evaluated with the standard `lerobot-eval`, so the test runs on any benchmark that LeRobot supports, and any VLA in LeRobot can serve as its key.

We apply the probe to nine public LIBERO fine-tunes [3, 4, 5, 6, 7, 8, 9, 10, 11] on LIBERO and on the perturbed scenes of LIBERO-plus [12].

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Related work. Retrieval has been used as a policy [13, 14, 15], to select training data [16, 17], and as context for agents and VLAs [18, 19], where a plain nearest-neighbour replay already rivals generalist agents [18]; He et al. [14] argue that diffusion policies memorise an action lookup table. Perturbation benchmarks show that VLAs memorise action sequences and layouts [12, 20, 21], interpretability work finds episode-specific memorisation features [22] and separate roles for VLM and action pathways [23], and VLA fine-tuning is known to erode the VLM's vision-language skills [24, 25, 26]. We turn retrieval into an instrument: it measures how much of a benchmark is retrieval, scores a VLA's backbone *as an action-relevant key* by closed-loop success, before and after fine-tuning, and measures what its action head adds.

2 A kNN probe

The policy. The memory holds every second frame of the 1,693 LIBERO demonstrations (137k keys), each with the chunk of actions that followed. "Training" is embedding them. At test time the policy embeds the observation o with ϕ and executes

$$\pi(o) \;=\; \sum_{i \,\in\, \mathcal{N}_k(o)} w_i \, a_{t_i:t_i+H}, \qquad w_i \propto e^{-d(o,\,o_i)/\tau}, \tag{1}$$

over the k nearest demonstration frames o_i, at most one per demonstration. The distance d adds image distances (per camera), state, prompt (CLIP [27]), time-in-episode and history terms, each scaled by its median within the task. Two extras make the average safe: *mode seeking* down-weights neighbours whose chunks disagree with the dominant one instead of averaging them into an action no demonstrator took, and *temporal ensembling* blends overlapping chunks [28]. Every executed action is therefore a convex combination of demonstrated actions. An optional *retry* rewinds and retrieves again when the scene looks unlike any demonstration; it did not help and is off. Hyperparameters were tuned with Optuna [29] (Appendix A). With a frozen ImageNet ResNet-18 [30] key (layer 3, 4×4 grid per camera), this *full method* needs no training; the *plain lookup* drops every extra.

VLM keys. Any VLA's vision-language model can be the key ϕ: the hidden states of one layer over each camera's image tokens, standardised per dimension (VLM residual streams are dominated by a few massive activations [31]) and pooled to a 2×2 grid. The layer is chosen once per family by an offline pilot (held-out action R^2) and kept for the checkpoints before and after LIBERO fine-tuning.

Offline score. A lookup only uses the nearest neighbours, so we score a key by its local action consistency, LAC@8 = 1 − the action distance from a held-out frame to its 8 nearest frames of other demonstrations, divided by the distance to a random frame of the same task.

3 Setup

LIBERO [1]: 4 suites × 10 tasks, ∼42 demonstrations per task. Every policy runs the official protocol, 50 episodes per task (2,000; the full kNN method with `lerobot-eval`, the others in our evaluation harness on the same initial states). **LIBERO-plus** [12]: the same tasks under seven perturbations (camera, robot initial state, language, light, background, sensor noise, layout); a fixed sample of 420 of its 10,030 variants (60 per perturbation), the same for every policy, never trained on. **VLAs**: nine public LIBERO checkpoints from the LeRobot Hub [2], evaluated as released (Appendix B). Nothing is trained except the optional learnt key of §5.

4 How hard is the task?

Table 1 places the lookup among the VLAs. **A training-free lookup is competitive**: 91.7% on LIBERO, rank 7 of 10. **Under perturbation it degrades least**: 9.3 points, against 12.2 for the most robust VLA and up to 64.8; it ranks 3 and is the best policy on two suites. Part of this is a favourable benchmark: LIBERO-plus scenes need the same actions as the clean demonstrations, so the answer is in memory whenever the key ignores the perturbation.

	LIBERO					LIBERO-plus					
	avg	Spat	Obj	Goal	Long	avg	Spat	Obj	Goal	Long	drop
$\pi_{0.5}$	**97.7**	97	100	**98**	**96**	**85.5**	**90**	84	83	**85**	−12.2
MolmoAct2	97.5	**97**	99	98	96	82.6	90	82	76	83	−14.9
kNN, full method	91.7	88	100	98	81	82.4	79	**86**	**87**	78	**−9.3**
kNN, + learnt key	93.0	91	100	97	85	81.7	88	84	81	74	−11.3
GR00T N1.7	94.8	96	98	95	91	80.2	90	74	79	78	−14.6
kNN, plain	88.7	90	95	95	74	75.5	77	82	79	64	−13.2
VLA-JEPA	97.2	95	**100**	97	96	74.0	81	70	77	68	−23.1
Evo-1	95.4	92	99	97	94	71.4	74	73	70	69	−24.0
X-VLA	94.5	95	99	93	92	64.5	71	65	65	57	−30.0
π_0	71.9	70	86	86	45	42.9	54	42	47	29	−29.0
SmolVLA	82.2	79	92	88	70	27.9	30	50	11	20	−54.3
π_0-FAST	90.5	90	94	91	87	25.7	39	34	16	13	−64.8

Table 1: Success (%) on LIBERO (official protocol, 2,000 episodes) and on the same LIBERO-plus variants, ordered by LIBERO-plus. The kNN rows are training-free except the learnt key. Bold: best per column.

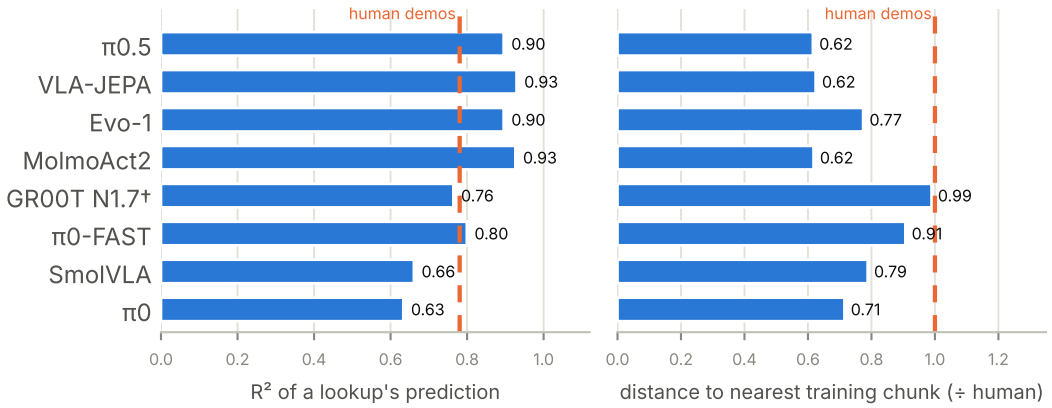


Figure 1: Clean LIBERO rollouts of each VLA. Left: R^2 of a lookup over the demonstrations (ResNet-18 key, $k = 8$) predicting the VLA's action chunk at the same state; dashed: the same for held-out human demonstrations. Right: the chunk's distance to the nearest training chunk, relative to a held-out human demonstration's. †Spatial suite. X-VLA predicts absolute poses and is left out.

VLAs fail where the lookup fails. When a VLA fails a LIBERO episode, the lookup fails the same episode in 20–46% of cases, against 8% overall (2.4–5.5× chance), and per-task success correlates with the lookup's (median $r = 0.74$; Appendix C). The weak VLAs key on the wording: reworded instructions over the original scenes drop SmolVLA, π_0-FAST and π_0 to 22, 33 and 12% (original wording: 82, 88, 68%).

5 Inside the VLAs

VLAs act like a lookup. At the states a VLA visits, a lookup over the demonstrations predicts its action chunk as well as it predicts a held-out human demonstration, or better ($R^2 = 0.90$ for $\pi_{0.5}$, 0.78 for humans; Fig. 1), and the VLA's chunks are closer to a training chunk than a new human demonstration is: they are not new. Off the demonstrations, on recorded LIBERO-plus rollouts, the robust VLAs stay lookup-like in their own key space ($R^2 = 0.85$ for $\pi_{0.5}$, 0.68 for VLA-JEPA), the collapsing ones do not (0.08 for SmolVLA, -0.01 for π_0-FAST).

Backbone and action head. A lookup keyed on a VLA's own VLM scores its backbone; the VLA's excess over it is its action head's (Table 2). *Some action heads subtract*: 2 of 9 VLAs score below a lookup on their own fine-tuned VLM on LIBERO, 4 of 9 under perturbation, by up to 48 points. GR00T froze its VLM during fine-tuning, so everything it learned on LIBERO is in its action head.

	LIBERO			LIBERO-plus		
	kNN before	**kNN** after	VLA	**kNN** before	**kNN** after	VLA
$\pi_{0.5}$	85.4	81.8	97.7	60.5	65.7	85.5
MolmoAct2	78.6	91.3	97.5	52.6	81.0	82.6
GR00T N1.7*	87.5	87.5	94.8	66.4	66.4	80.2
VLA-JEPA	84.0	83.0	97.2	49.3	65.7	74.0
Evo-1	62.1	87.1	95.4	35.5	**72.6**	71.4
X-VLA	83.3	88.5	94.5	62.4	**78.1**	64.5
π_0	86.9	17.9	71.9	56.7	2.4	42.9
SmolVLA	76.3	**86.5**	82.2	45.0	**69.3**	27.9
π_0-FAST	86.6	**95.0**	90.5	58.3	**74.0**	25.7

Table 2: A plain lookup keyed on each VLA's own VLM before and after its LIBERO fine-tuning, and the VLA itself (success, %). Orange: the lookup on the fine-tuned VLM beats the VLA. *GR00T froze its VLM, so before = after.

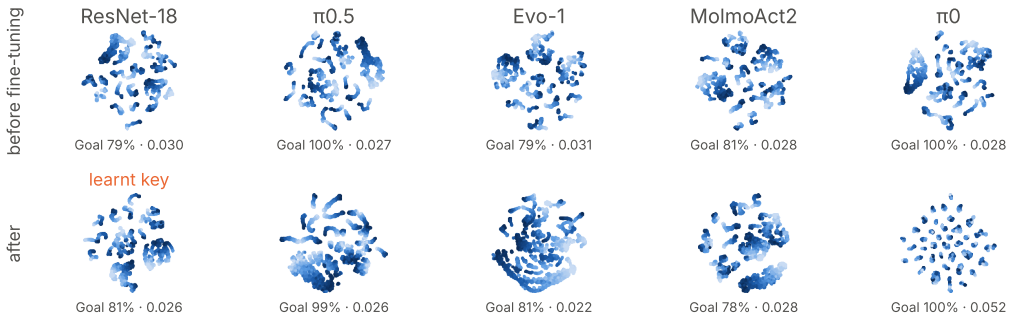


Figure 2: t-SNE of the lookup keys of the same 4,150 demonstration frames, coloured by progress through the episode (light: start, dark: end), before (top) and after (bottom) LIBERO fine-tuning. Below each map: same-task nearest neighbours from other demonstrations on LIBERO-Goal (one scene, ten instructions) and the median progress gap to them.

What fine-tuning does to the backbone. It mostly makes the VLM a more robust key: +5 to +37 points on LIBERO-plus for 7 of 8 VLAs, while clean success improves for only 5. Maps of the keys show what changes (Fig. 2, Appendix D): the scene stops dominating them, and they track the phase of the task (the progress gap shrinks for 7 of 8 VLMs). Fine-tuning can also break the backbone: π_0's key still tells the tasks apart but loses the phase ($0.028 \rightarrow 0.052$), and its lookup collapses ($86.9 \rightarrow 17.9\%$).

Scoring a backbone without rollouts. LAC tracks the closed-loop success of a lookup with each key (Pearson $r = 0.78$, Spearman $\rho = 0.61$ over the 12 keys that never saw these demonstrations; Appendix C) and flags a broken backbone: π_0's fine-tuned key has LAC 0.55, against 0.76–0.88 for every other key. Fine-tuned VLMs score high on their own training demonstrations, so LAC on them flags memorisation. As a loss (one linear map per ResNet grid cell) it lifts the plain lookup $88.7 \rightarrow 93.0\%$ and the full method to 93.0%.

6 Limitations and conclusion

LIBERO has dense demonstrations and test scenes close to them, the regime where retrieval works; our LIBERO-plus numbers come from a 420-variant sample (about ± 3.5 points overall), and its perturbed scenes keep the demonstrated actions valid. A lookup cannot extrapolate beyond its memory: we propose it not for robot control but as a cheap, training-free test to report next to every VLA result on a dense-demonstration benchmark (how much of it is retrieval, how good the backbone is and what fine-tuning did to it, what the action head adds), on any benchmark LeRobot supports.

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A The kNN policy: tuning and ablations

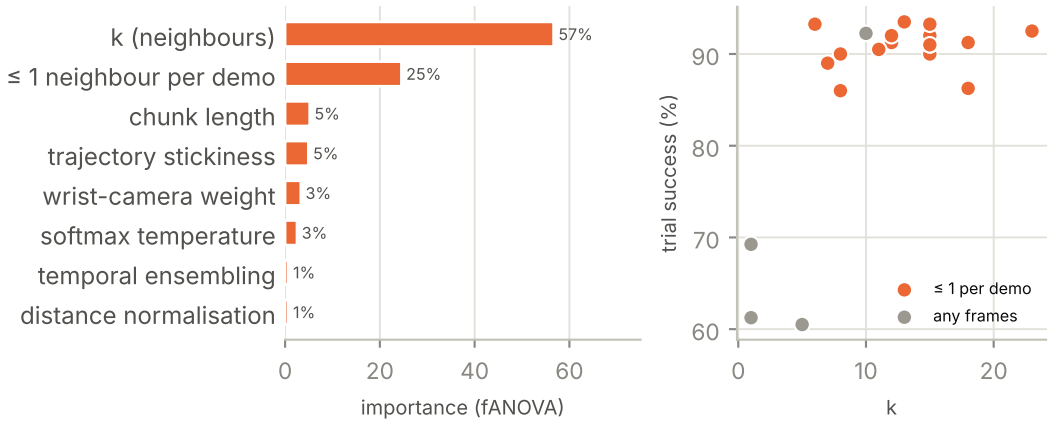


Figure 3: Hyper-parameter search (Optuna, 24 trials of 400 episodes in the final study, 193 over all studies, on initial states 40–49 and 0–19). Left: fANOVA importance [32]. Right: every trial's success against k. LIBERO has no held-out split, so the official evaluation includes these states, as for every LIBERO result.

Change to the full method	avg	Spatial	Object	Goal	Long	Δ
none (full method, `lerobot-eval`)	91.7	88.0	99.8	97.6	81.4	
prompt ignored	64.6	77.6	99.6	8.8	72.4	−27.1
raw 16×16 pixels	57.1	79.0	0.0	91.0	58.6	−34.6
10% of demos	77.4	83.8	87.0	90.2	48.6	−14.3
25% of demos	82.6	83.6	97.8	92.6	56.4	−9.1
50% of demos	88.5	86.4	99.6	95.4	72.6	−3.2
k = 1 (replay nearest demo)	78.6	77.2	90.6	86.2	60.4	−13.1
DINOv2-S instead of ResNet-18	83.4	79.8	97.6	95.0	61.0	−8.3
agent-view camera only	82.0	78.0	92.8	93.6	63.8	−9.7
wrist camera only	87.5	87.0	98.8	92.8	71.2	−4.2
plain weighted mean	90.7	88.4	100.0	96.6	77.8	−1.0
no temporal ensembling	91.1	92.0	98.0	94.2	80.2	−0.6
no history matching	91.5	89.6	98.8	97.6	79.8	−0.2
no proprioceptive state	91.3	87.6	99.6	96.8	81.4	−0.4
no time-step feature	91.3	88.2	99.8	97.4	79.8	−0.4
hard task filter	91.9	88.4	100.0	97.4	81.8	+0.2
k = 2 (13 in the full method)	85.2	83.8	94.6	92.2	70.2	−6.5
k = 4 (13 in the full method)	89.3	86.2	98.8	93.8	78.6	−2.4
k = 8 (13 in the full method)	89.8	85.4	99.6	96.6	77.6	−1.9
k = 24 (13 in the full method)	92.7	89.2	99.8	97.2	84.4	+1.0
k = 32 (13 in the full method)	91.7	88.8	99.8	97.4	80.8	+0.0

Table 3: One change to the full method at a time, official protocol (2,000 episodes). Δ: against the full method. $k = 24$ scores slightly above the tuned $k = 13$, within the ±1.2-point 95% interval of 2,000 episodes. Raw pixels fail all of LIBERO-Object: at 16×16 its ten tasks share a scene of small objects.

B Evaluated VLAs

All checkpoints are public and evaluated as released, at their LeRobot settings: $\pi_{0.5}$, π_0 and π_0-FAST (LeRobot LIBERO fine-tunes), SmolVLA, X-VLA (absolute end-effector control and its own observation convention), VLA-JEPA (7 actions per re-plan), Evo-1 (14 actions per re-plan; 256 px renders on both benchmarks, as LIBERO-plus noise assumes them), MolmoAct2 (continuous actions), GR00T N1.7 (one checkpoint per suite, 8 of 16 actions per re-plan). Every checkpoint first

passes an offline check: on demonstration frames its predicted motion must be closer to the demonstrated one than half the error of a shuffled pairing.

C More results

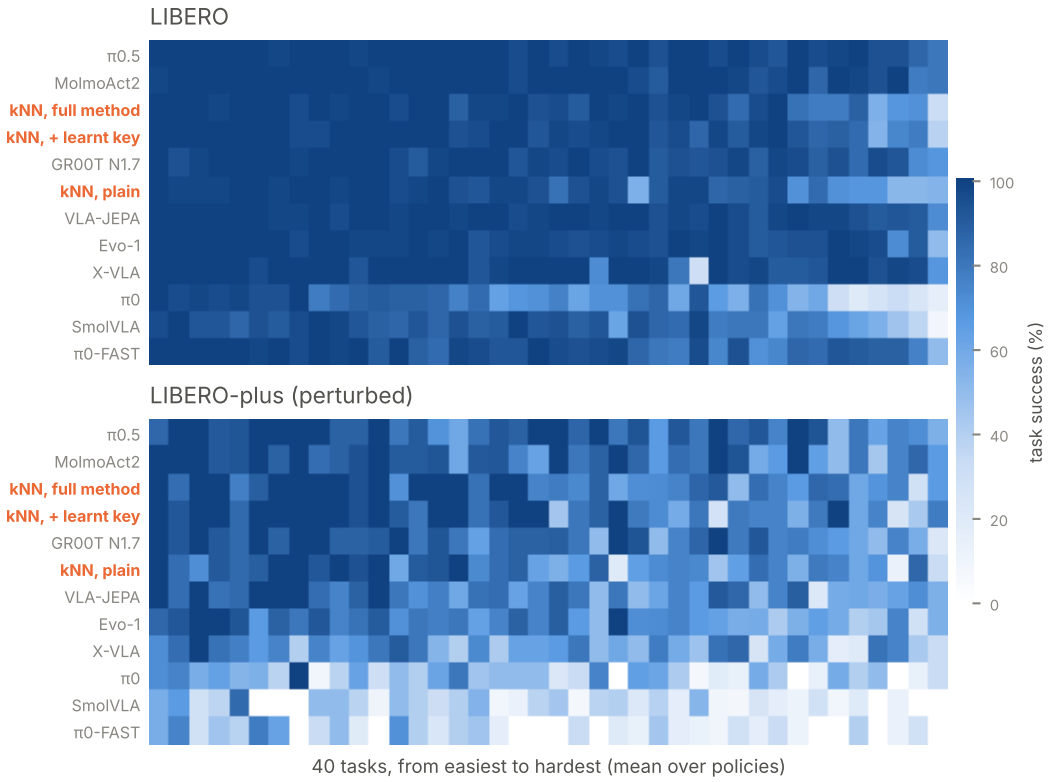


Figure 4: Success of every policy on each of the 40 tasks, tasks sorted from easiest to hardest (mean over policies).

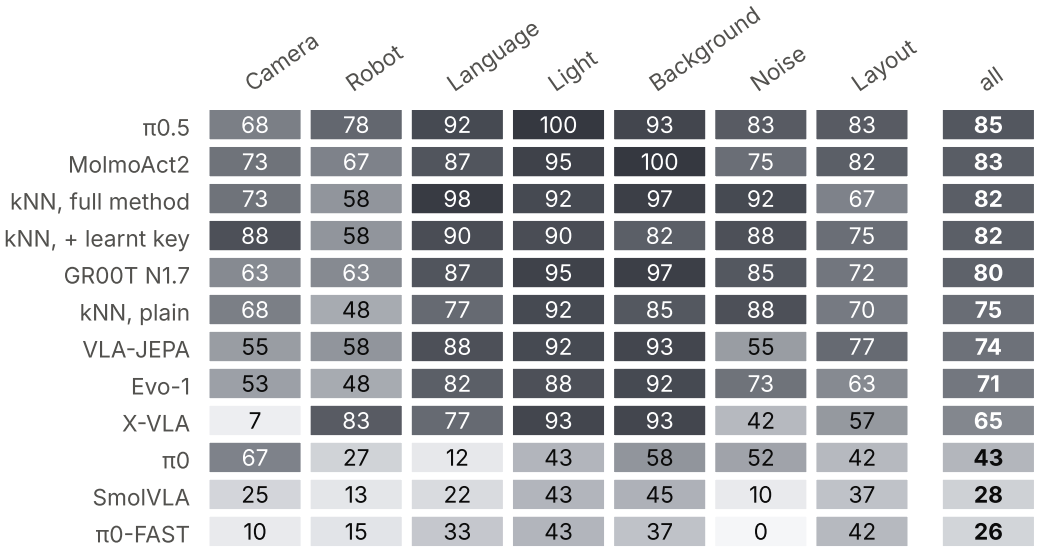


Figure 5: LIBERO-plus success (%) per perturbation, same variants for every policy.

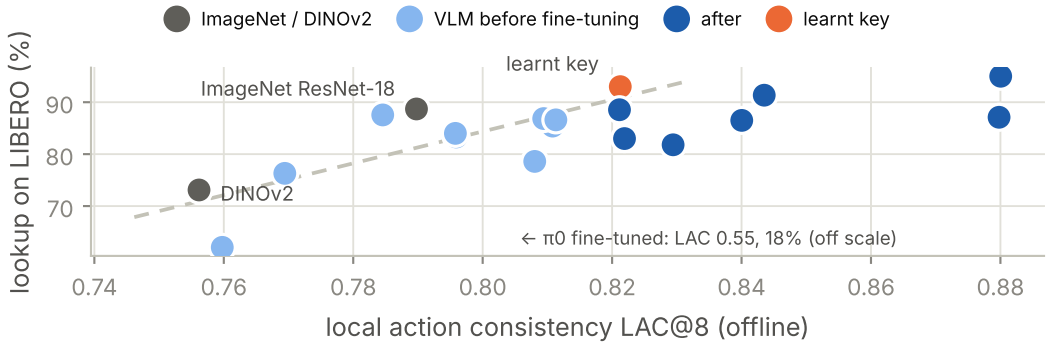


Figure 6: Local action consistency of a key (offline, held-out queries) against the LIBERO success of a plain lookup with that key; one point per backbone key, before and after LIBERO fine-tuning.

D Embedding maps

Every key of Table 2, plus the generic keys, on the same 4,150 demonstration frames (6 random demonstrations per task, every 10th frame). Each key is computed as the plain lookup computes it: standardised with memory statistics, unit-normalised per camera, cameras concatenated. Fig. 7: t-SNE (perplexity 30, on the first 50 principal components), coloured by progress through the episode. Fig. 8: the first two principal components, coloured by suite. Before fine-tuning the suites, i.e. the scenes, dominate every key (the suite explains 40–84% of the variance of the two components; ImageNet ResNet-18: 85%); after it far less (1–43%, lower for 8 of 8 VLMs), consistent with the fine-tuned keys' robustness to scene perturbations. Only keys whose image tokens attend to the instruction, under the bidirectional prefix attention of PaliGemma (π_0, $\pi_{0.5}$, π_0-FAST) and of Florence-2's encoder (X-VLA), tell LIBERO-Goal's ten tasks apart: 96–100% same-task neighbours, against 76–82% for the VLMs with causal attention, whose image tokens precede the instruction, and 79% for ResNet-18.

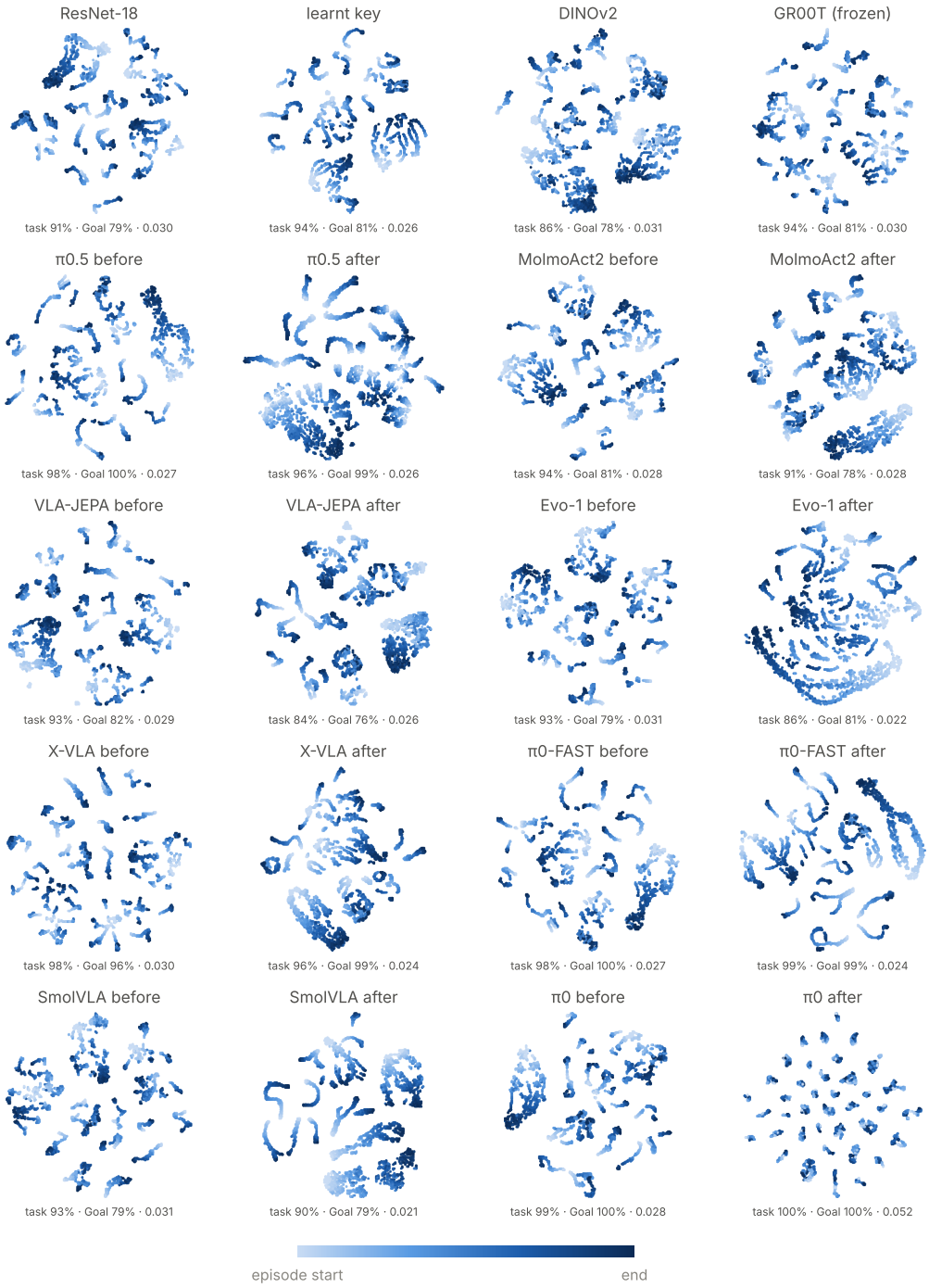


Figure 7: t-SNE of every key, coloured by progress through the episode. Below each map: same-task nearest neighbours from other demonstrations (all tasks; LIBERO-Goal) and the median progress gap to the nearest frame of another demonstration of the same task. GR00T froze its VLM, so its key is the same before and after fine-tuning.

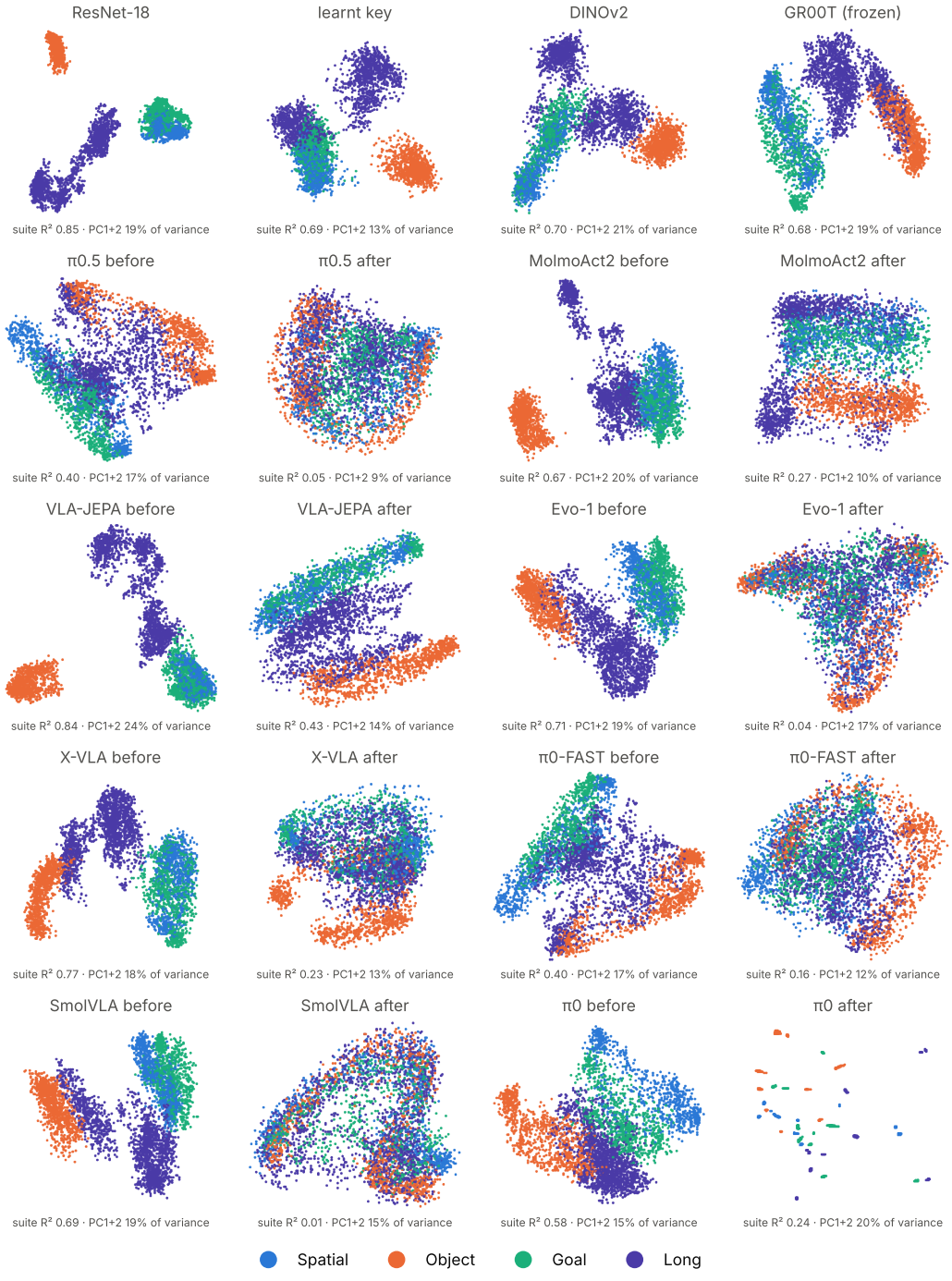


Figure 8: The first two principal components of every key, coloured by suite. Below each map: the share of their variance the suite explains, and the share of the key's variance they explain.